INTERNAL STUDY ON OUTBOUND LOGISTICS AND 3PL COORDINATION AT ASHOK LEYLAND

An Analytical Evaluation Across Location with Strategic Recommendations

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Executive Summary

This report presents a comprehensive, data-driven analysis of the outbound logistics operations at Ashok Leyland, with a specific focus on the performance and coordination challenges faced in managing third-party logistics (3PL) providers. The study covers 9 key locations and synthesizes responses from over 50 internal logistics professionals across execution, planning, and management roles.

The primary objective was to uncover operational frictions, evaluate digital readiness, and assess the structural robustness of Ashok Leyland's 3PL engagement model. Using a structured questionnaire across ten challenge domains, the study provides clear, quantified insights into recurring bottlenecks, depot-level discrepancies, and scope for future growth and evolution.

Key findings reveal consistent pain points in driver management, digital coordination, SOP compliance, emergency dispatch cost control, and route planning. Issues such as dependency on informal communication (Q24), weak pre-planning by vendors (Q13), and elevated emergency freight costs (Q36, Q38) are especially prominent. Depot-level disparities were also significant, with Durgapur, Alwar and Pune facing relatively higher operational stress, while Hosur and Ennore demonstrated stronger baseline performance.

The strategic recommendations provided include:

- Creation of a centralised 3PL control tower
- Pilot testing hybrid or partial in-house logistics
- Deployment of route simulation tools, possibly even involving AI and fatigue management protocols
- Vendor-linked digital dashboards and in-house dashboard development as well as SOP training programs
- Introduction of micro-warehousing or last-mile solutions for remote zones, or maintaining a fleet of specialised drivers.

In addition, depot-specific interventions have been designed to reflect ground realities and provide tailored action plans in accordance to each plant location.

While the report provides actionable insights, it also acknowledges several limitations, including uneven respondent distribution across depots, the exclusion of vendor-side perspectives, and the constraints of a static quantitative method. These are critical considerations for future studies aiming to capture broader system-level dynamics and go into a much greater depth in order to gain more meaningful recommendations.

Overall, this internal study serves as both a diagnostic and strategic compass for Ashok Leyland's logistics leadership. It offers an empirical foundation for operational redesign, partner development, and future logistics transformation efforts across the company's outbound value chain.

Chapter 1: Introduction

1.1 Background and Industry Context

The Indian heavy commercial vehicle sector operates under increasingly complex logistics environments and dynamics, especially during outbound movements from OEM plants to regional dealerships, export points and even stock transfers. With growing pressure on delivery reliability, compliance, and customer satisfaction, the role of Third-Party Logistics (3PL) providers has become significantly grown and has become both indispensable and strategically critical.

Ashok Leyland, being one of India's largest CV manufacturers, relies extensively on its 3PL vendors for outbound vehicle distribution across its multi-plant footprint. These logistics partners handle the end-to-end movement of finished vehicles from plants such as Ennore, Hosur, Alwar, Pantnagar, and others. However, the scale at which it operates, fragmentation, and on-ground execution challenges across zones have highlighted several pain points in areas such as digital integration, SOP adherence, performance governance, and transit delays.

This study was initiated to conduct a systematic internal assessment of the 3PL coordination landscape, in order to understand how it functions, where it struggles, and how Ashok Leyland can build more resilience, efficiency, and accountability in its overall outbound supply chain.

1.2 Purpose of the Study

The core intent of this report is to assess the real-world operational challenges that are experiences, gaps in coordination, and digital readiness of the third-party logistics partners that are working with Ashok Leyland. While previous anecdotal inputs and incident-based reviews have pointed to recurring problems, this project attempts to create a data-backed understanding of the 3PL ecosystem from both a quantitative and analytical framework.

In particular, the study aims to identify:

- Recurring friction points in the day-to-day outbound operations
- Variability in vendor performance and coordination across different plants

- The extent of digital adoption and real-time visibility gaps that may exist
- Perceptions around SOP enforcement, driver compliance, and risk hotspots
- Comparative views on centralised coordination, hybrid models, and the potential for a 4PL architecture in the future

1.3 Strategic Relevance to Ashok Leyland

This study aligns closely with Ashok Leyland's aim to integrate broader logistics transformation efforts. As the industry moves toward a more digitally enabled era as well as centrally coordinated logistics frameworks, insights will thereby help stakeholders to:

- Help prioritise regions and pain points that require process redesign or digital tooling
- Provide a baseline for KPI benchmarking across 3PL vendors
- Enable leadership to make informed decisions about centralised logistics governance, technology investment, and 3PL partner development
- Inform the possible transition toward hybrid or 4PL models that promise integrated visibility and accountability

More than just a scorecard, the report is designed to serve as a diagnostic and strategic guide for the logistics function, which captures internal voices from across zones and experience levels to reflect the operational realities that are experienced on the ground.

1.4 Scope and Coverage

This internal analysis focuses solely on outbound logistics operations involving third-party logistics providers. It covers:

- Quantitative survey responses from over 50+ logistics professionals across key plant locations
- Experience-based insights ranging from mid-level to senior managers
- Assessment across 10 core challenge domains, including SOPs, transit risk, digital coordination, performance governance, and future models
- Plant-level comparisons, correlation with satisfaction metrics, and perception-driven conclusions that will enable deeper understanding

Inbound logistics, part-load movement, and plant-level inventory operations are out of scope for this study.

1.5 Intended Audience

This report is designed to serve the needs of:

- Corporate Logistics and Supply Chain Leadership
- Plant-level Logistics Managers and Planning Teams
- Digital Transformation and Systems Teams
- Vendor Management & Procurement Stakeholders

Chapter 2: Industry Overview of Outbound Logistics in Heavy Vehicle Manufacturing

2.1 The Outbound Logistics Landscape

Outbound logistics in heavy vehicle manufacturing involves transporting bulky, high-value vehicles, that include trucks and buses, these involve routes from factory gates to regional depots, dealers, or ports. These vehicles typically require:

- Multi-axle trailers,
- Coordination with yard teams for loading/unloading,
- Regulatory permits for axle loads and inter-state transit.

This contrasts with lighter goods supply chains and underscores the complexity OEMs like Ashok Leyland have to manage and may have to also delegate.

2.2 Role and Rise of 3PLs

Third-party logistics (3PL) firms have become central to managing this complexity:

- They offer flexibility and scale, allowing OEMs to avoid owning and maintaining their own transport fleets and driver pools which are capital-intensive and underutilised outside the peak dispatch times
- They bring technical expertise and specialisation in route planning, permit procurement, and yard handling at stockyards and ports .
- The Indian 3PL market, valued at approximately US $42–43 billion in 2025, is growing above 7% annually and plays a significant role in the automotive outbound chains

For OEMs like Ashok Leyland, 3PL deployment has become standard due to:

1. Complexity of vehicle handling,
2. Need for decentralisation that are dispersed across geographies,
3. Intermittent volume cycles which are aligned with production schedules.

2.3 Challenges in the Indian Industry

Outside the factory gate lies a highly fragmented logistics landscape, bringing several common issues:

- High logistics costs: Roughly 13–14% of India's GDP which are driven by outdated systems and poor road infrastructure.
- Fragmented vendor networks: Many small-scale 3PL/vendors create variability in digital capabilities, compliance checks, and SOP discipline.

- Underuse of digital tools: Estimates suggest only ~15% of 3PLs use advanced digital tools like GPS, telematics, or analytics thereby limiting visibility and timely response .

These factors often fold out as delayed deliveries, inconsistent SOP adherence, high emergency freight expenses, and poor governance of driver and transit practices.

2.4 Emergence of 4PL and Its Relevance

Fourth-party logistics (4PL) or Lead Logistics Integrators (LLIs) have thereby emerged to address these fragmentation issues. A 4PL:

- Acts as a single point of accountability that would effectively orchestrates multiple 3PLs,
- Offers end-to-end visibility, data integration, and analytical insights across transportation, warehousing, and compliance
- Transitions the relationship from tactical execution to strategic orchestration and network optimisation.

This shift is becoming visible in industries with complex outbound flows, like heavy commercial vehicles wherein the stakeholders seek improved vendor coherence, performance governance, and digital oversight.

2.5 Strategic Benchmarking: Ashok Leyland

In comparison to its peers:

- Tata Motors, M&M, and Volvo-Eicher have begun trials with e-Logistics control towers or hybrid 4PL pilots in order to improve visibility and reduce inefficiency in their processes.
- Ashok Leyland's continued investment in visible systems such as eRFID-assisted gate processes, also signifies readiness to evolve its logistics capabilities.

By situating the company within this broader industry trajectory, your report justifies the data-driven recommendations around centralised governance, digital rollout, and the exploration of hybrid/4PL models.

2.6 Conclusion

The heavy vehicle manufacturing sector in India is currently at a logistics inflection point. Rising 3PL dependence meets systemic fragmentation, prompting OEMs to evolve toward more integrated models. In this context, this report's findings and recommendations would take on a greatly added relevance which is to identify Ashok Leyland 3PL capabilities and in order to understand pain points that may arise from this particular scope across various locations of the company.

Chapter 3: Research Methodology

3.1 Research Purpose and Question:

The purpose of this research is to identify and evaluate the most common challenges that are faced by the logistics and supply chain managers that work at Ashok Leyland when working with the third-party logistics (3PL) service providers. Ashok Leyland being one of India's largest commercial vehicle manufacturers, relies on a complex 3PL network to support both the inbound supply components as well as the outbound delivery of finished vehicles. Therefore, understanding the internal perspective on 3PL coordination as well as challenges will thereby help improve logistics effectiveness and partner relationships.

The research question through which we want to lead this thesis is as follows:

"What are the most common challenges reported by heavy automotive logistical managers when working with 3PLs?"

3.2 Research Design:

This study adopts a quantitative, descriptive survey design. This will be characterised by a structured questionnaire which will be used to collect numerical data from a sample of logistics professionals that are associated with such cases. This approach will therefore allow for:

1) Measuring the prevalence of specific challenges that exist.

2) Identifying trends across regions and logistics types for both the inbound and the outbound aspects.

3) Ranking the severity of the issues that are experienced.

3.3 Target Population and Sampling:

Target Respondents include:

1) Logistics or supply chain managers that work in a large and established OEM such as Ashok Leyland.

Sampling Technique:

1) Purposive sampling will be used to reach respondents that hold relevant logistics experience.
2) A target sample size of 50 to 100 respondents will be apt to create a meaningful statistical survey with fruitful insights.

3.4 Data Collection Procedure:

The survey will be distributed online through channels such as Google Forms or Microsoft Forms, as per the convenience of the respondents.
Pilot tests may be considered with a few industry professionals to validate genuinity, reliability and even garner suggestions from them, in order to help improve the study.

3.5 Data Analysis Plan:

Data will be analysed using tools such as Microsoft Excel and SPSS. Some techniques that would be used are as follows:
Descriptive Statistics: Frequencies, percentages, and mean scores.
Weighted Ranking: To identify the most significant challenges across respondents.
Cross-Tabulation: For comparing challenges that prevail across different variables such as region, firm size and many more.
Charts and Pictorial Representations: Use of bar graphs, pie charts, heat maps and others will help in generating clear and easily understandable visual representation.

3.6 Ethical Considerations:

Participation will be voluntary and anonymous.
Respondents will also receive a clear consent statement, which will enable them to express content before beginning a survey.
No identifiable information will be collected and shared.
All data will be stored securely and used solely for the purpose of academic research.

Chapter 4: Data Analysis

4.1 Respondents Overview

4.1.1 Functional Representation and Focus

A majority of respondents (80.39%) are directly involved in outbound logistics, with only 9 respondents representing inbound and 1 managing both. This validates the outbound-centric orientation of the study and ensures the findings are strictly relevant to the subject of vehicle dispatch and aspects related to it.

4.1.2 Location Distribution

The most represented locations are mainly Ennore (21.57%), Pantnagar (17.65%), Hosur (17.65%), and ALCOB (15.69%). Smaller depots like Durgapur, Bhandara, and Vijayawada are only characterised by a few participants, which may slightly restrict generalisability of the analysis to all the locations but still captures a broad operational footprint from whatever resources were at our disposal.

4.1.3 Experience and Educational Background

The logistics workforce in this study is fairly experienced:

- Over 70% have more than 2 years of experience
- Notably, 29.41% have 10+ years, adding credibility to their assessment of long-term 3PL performance, as compared to the past. Further, their analysis could even be more deep.

From an academic standpoint, 58.82% hold a diploma, while a smaller portion has bachelor's (23.53%) or master's degrees (17.65%). This mix could also be attributed to the technical-operational nature of the logistics functions within the company.

4.1.4 Type of 3PL Partners

The respondents primarily engage with a large group of national 3PL firms (68.63%), followed by a smaller share dealing with regional SMEs or both. This reinforces the relevance of the findings to institutional-scale transporters and their systemic performance as well.

4.1.5 Satisfaction with 3PL Providers

Based on Q7:

- A significant 84.3% rate satisfaction between 3 and 4, with no one rating it as low as 1 (not at all satisfied).
- The average satisfaction score is 3.57, median is 4, and standard deviation is 0.76, thereby indicating a fairly consistent perception of acceptable but also showcasing a scope to improve.
- Only 4 respondents gave a high rating of 5.

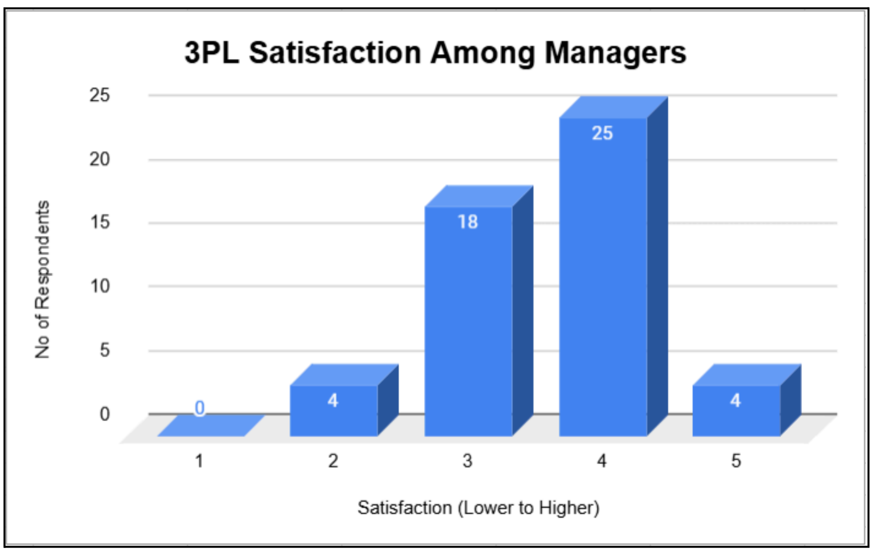

Figure 4.1: 3PL Satisfaction Among Managers

This suggests that while extreme dissatisfaction is absent, there is a moderate level of concern or underwhelming service that needs to be examined in the deeper operational and strategic areas, in order to gain even more overall efficiency.

4.2: Common Operational Challenges

4.2.1 Overall Assessment of Operational Challenges

The overall mean score for operational challenges (Q9–Q19) is 2.80, indicating a slightly moderate level of operational friction that can be found in Ashok Leyland's outbound logistics processes. Scores nearing 5 suggest very severe issues, while lower values imply smoother operations. The current mean reflects consistent yet manageable issues requiring periodic oversight and intervention from a logistics standpoint.

4.2.2 Challenges Requiring Immediate Attention (High Scores)

The most prominent operational hurdles, as highlighted by the highest average scores, are:

- Inbound Delivery Delays (Q9: Mean = 2.92): Slight delays experienced on the inbound side could possibly affect downstream outbound coordination. This serves as a weak link that could ripple across operations as well.
- Outbound Delivery Timeliness (Q10: Mean = 2.90): Slight delays in dispatch adherence could compromise customer satisfaction and challenge internal resource planning as well. The issue could persist as a foundational challenge in this department.
- Lack of Proactive Route Planning by 3PLs (Q13: Mean = 2.98): Suggests a reactive stance from transport vendors, reflecting slight lapses in pre-planning, which may reflect in avoidable inefficiencies.
- Driver Non-compliance with Routes and SOPs (Q15: Mean = 2.82): Deviation from prescribed practices is flagged as a recurring minor issue, often due to lapses in driver monitoring or inconsistent training practices.
- Rapid Resolution of Disruptions (Q17: Mean = 2.9): Disruption handling is also above the moderate lines, hence pushing for more opportunities in order to improve turnaround time during exceptions in order to push for operational excellence.

4.2.3 Challenges Moderately Affecting Operations (Intermediate Scores)

These challenges are not as prominent but are still regularly experienced across depots:

- Failure to Accommodate Urgent Requests (Q16: Mean = 2.80): This could indicate mixed flexibility from transport partners, particularly when they are handling last-minute or escalated delivery requirements.
- Transit Damages and Losses (Q14: Mean = 2.76): Slight but notable concern suggesting inconsistencies in vehicle condition, loading practices, or driving behaviour that may impact asset integrity and delay the movement of vehicle to the end customer.
- Communication Inconsistencies with 3PL Teams (Q11: Mean = 2.69): Ongoing communication gaps which may occasionally hinder coordination between Ashok Leyland staff and the external logistics operators that they have employed.

4.2.4 Areas with Relatively Fewer Challenges (Low Scores)

These dimensions indicate comparatively fewer issues and reflect areas where performance is relatively stronger within the department:

- SOP Compliance by 3PLs (Q18: Mean = 2.51): Implies that procedural compliance from third-party logistics partners is largely being maintained and implemented.
- Clarity in Responsibility Ownership (Q19: Mean = 2.59): Respondents feel that escalation roles and ownership are generally well understood and reasonably enforced within the entire ecosystem.

4.2.5 Relationship between Challenges and Overall Satisfaction

Several items demonstrate notable correlations with overall satisfaction (Q7), highlighting their role in shaping vendor perception:

- Clarity in Responsibility Ownership (Q19: correlation = +0.18): Shows a moderately positive impact on satisfaction, implying that clearer accountability builds greater confidence within the logistics network.
- SOP Compliance (Q18: correlation = +0.31): Reflects a strong positive correlation, reinforcing that strict adherence to procedures by 3PLs directly influences stakeholder satisfaction.
- Communication Gaps (Q11: correlation = -0.22) and Outbound Delivery Timeliness (Q10: correlation = -0.19):

Both continue to show negative associations, confirming that inconsistent updates and missed schedules significantly erode trust in these related services.

4.2.6 Depot-level Comparative Insights

The depot-wise total average scores highlight regional variations in challenge intensity:

High-Challenge Depots:

- Durgapur (Mean = 3.54): This depot continues to report severe challenges across nearly all items, especially delivery timeliness and coordination, warranting targeted troubleshooting and possible vendor restructuring, although we must note that there is only a solitary respondent representing them.
- Alwar (Mean = 3.12), Bhandara (Mean = 2.9) and Pantnagar (Mean = 2.94): Indicate moderate challenge levels as well and are primarily highlighted by inconsistencies in SOP adherence and communication gaps.

Moderate-Challenge Depots:

- Hosur (Mean = 2.78) and Ennore (Mean = 2.63): Reflect issues that are persistent but manageable. Common challenges revolve around compliance, proactive route planning, and driver discipline which can be resolved to further strengthen processes.

Low-Challenge Depots:

- Vijayawada (Mean = 2.14) and ALCOB (Mean = 2.61): Stand out as locations which claim relatively strong performance. Effective coordination, discipline in SOP compliance, and fewer delivery delays contribute to these locations showing operational maturity.

4.3: Technology and Digital Adoption

4.3.1 Overall Digital Maturity of 3PL Partners

The mean rating for digital maturity among 3PL partners (Q21) is 3.31, reflecting moderately positive digital adoption. While approximately 41.2% of respondents rated maturity as average (3) and 31.4% as good (4), a limited proportion of respondents (9.8%) recognized high maturity (5). Thus, there's clear potential for enhancing digital capabilities among the 3PL providers for greater efficiency and ease of operations. It is also important to note that this question was not reverse coded, so the higher the value the better it is.

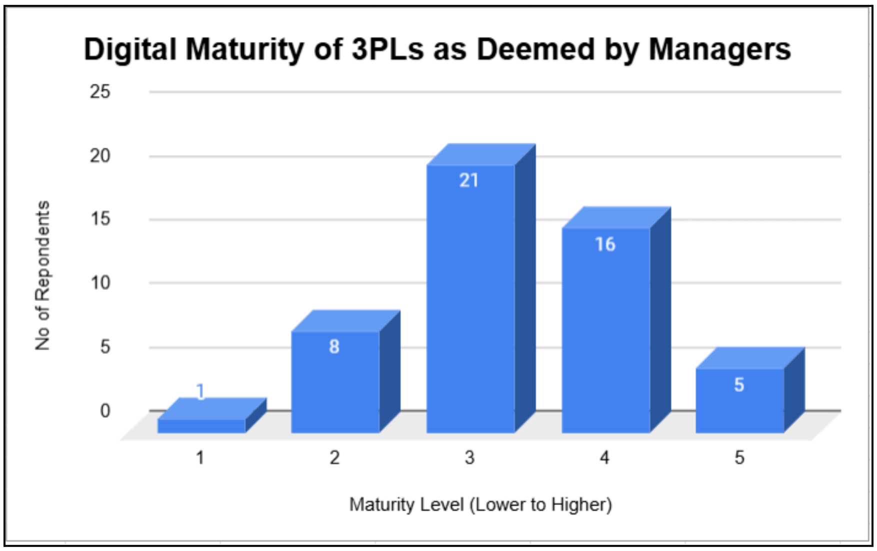

Figure 4.2: Digital Maturity of 3PLs as Deemed by Managers

4.3.2 Specific Technological Challenges and Strengths

Areas with relatively higher technological challenges (higher scores):

- Dependence on informal communication (Q24: Mean = 3.63): Strong reliance on informal methods (phone or WhatsApp) rather than structured digital systems highlights inconsistent and loose communication which could potentially lead to risks of miscommunication or loss of information.

- Lack of digital tools affecting quick decision-making (Q23: Mean = 3.69): Indicates that gaps in technology availability are clearly impacting responsiveness and decision efficiency which could lead to operational delays overall.

Areas with fewer technological challenges (lower scores):

- Real-time updates improving operational efficiency (Q22, reverse-coded Mean = 2.16): Suggesting real-time updates are broadly available and effective in aiding managers to make favourable operational decisions.
- Digital dashboard usage (Q26, reverse-coded Mean = 1.96): Relatively effective deployment of digital dashboards indicates satisfactory transparency in operational metrics tracking which help in the overall process.

4.3.3 Correlation with Overall Satisfaction

Correlations are modest but highlight key areas needed for management satisfaction:

- Digital maturity (Q21) has a positive correlation (0.42) with satisfaction, strongly indicating that greater digital maturity of partners significantly improves overall satisfaction in order to confidently carry out operational activities.
- Real-time updates (Q22, reverse-coded correlation = -0.07) show minor negative correlation, suggesting that while helpful, their impact alone on satisfaction is less pronounced without broader digital integration, which may be due to demand for greater digital tools to make more meaningful decisions and in some cases to reach meaningful conclusions.

4.3.4 Depot-level Analysis of Digital Adoption and Challenges

- Depots demonstrating higher technological challenges:

    - Ennore (Avg. Score = 2.73) and ALCOB (Avg. Score = 2.83) indicate moderate challenges, especially regarding dependence on informal communications and limited structured and consistent digital interactions.

- Durgapur (Avg. Score = 3.2) and Vijayawada (Avg. Score = 3) and Alwar (Avg. Score = 2.71), showed issues with regards to real-time updates and information relay through informal channels such as Whatsapp.
- Hosur (Avg. Score = 2.75) and Pune (Avg. Score = 3.1), showed concerns regarding the use of the digital tools as well as using informal communication systems for updates.

- Depots with relatively fewer technological challenges:

  - Bhandara (Avg. Score = 2.4) relatively show high level of challenges, suggesting more stability as compared to the other locations.

4.3.5 Technological Tools Currently in Use (by depot and experience)

- GPS tracking is nearly universal across depots and experience levels, reflecting how essential it is to track the vehicle.
- Advanced tools such as Transport Management Systems (TMS) and Telematics are employed selectively:

  - ALCOB and Hosur exhibit comprehensive adoption across most experience groups.
  - Pantnagar and Ennore report mixed usage, often relying more on basic systems such as manual records or GPS only.
  - Smaller depots (Pune, Vijayawada, Bhandara, Durgapur) generally rely more heavily on manual methods as inferred from their responses thereby indicating lower digital maturity or infrastructure challenges in these regions.

4.4: Performance Measurement & Review

4.4.1 Overall Performance Measurement Practices

The mean score for performance measurement and review-related challenges (Q27–Q33) is 2.22, indicating relatively lower overall issues in performance management and review systems at place in the organisation. This suggests the presence of a good amount of practices which are present to regularly measure and manage the 3PL partner performance.

4.4.2 Specific Performance Measurement Challenges (Higher Scores)

- 3PL performance exceeding expectations (Q33, Mean = 2.43): Scoring the highest mean in this section, which shows that even though the service is deemed satisfactory changes can be done in order to make the whole process more effective and efficient.
- Structured reviews for performance issues (Q28, Mean = 2.31): Moderate scores point to challenges in addressing performance issues through structured reviews, indicating occasional gaps in the feedback mechanism and a small scope to strengthen the performance appraisal process.
- Feedback mechanism for reporting failures (Q29, Mean = 2.41): Indicates moderate gaps which may exist and can be used to escalate 3PL performance issues for better operational efficiency.

4.4.3 Areas of Strength in Performance Measurement (Lower Scores)

- Regular reviews based on clear KPIs (Q27, Mean = 2.10): The low score here indicates fewer challenges, suggesting relatively decent adherence to regular KPI-based reviews, although there may be a scope to widen the KPI-base and identify more challenges.
- Tracking and sharing delivery performance metrics (Q30, Mean = 2.02): Indicates a good, established practice of monitoring and sharing performance data internally and possibly with the 3PLs as well.
- Penalties enforced for repeated failures (Q31, Mean = 2.20) and Measurable improvements in 3PL performance (Q32, Mean = 2.10): These relatively low scores

imply a good and effective penalty mechanisms in place which result in positive shifts in 3PL provider performance over the past two years.

4.4.4 Correlation with Overall Satisfaction (Q7)

Correlations are weak across the board, indicating that the operational aspects measured here may not strongly influence overall satisfaction. Notably:

- Tracking and sharing delivery performance (Q30, correlation = 0.05) shows minimal impact on overall satisfaction, suggesting that basic performance tracking is effectively established in place and no longer significantly influences the perception of satisfaction.
- Penalties enforced (Q31, correlation = 0.14) and feedback mechanism (Q29, correlation = 0.05) also demonstrate really minimal but positive impacts, indicating slight improvements in these areas might marginally enhance satisfaction.

4.4.5 Depot-Level Analysis of Performance Practices

- Ennore (Mean = 1.92) and Hosur (Mean = 2.17) demonstrate notably fewer performance measurement challenges, indicating relatively stronger local performance review practices and feedback mechanisms that are in place.
- ALCOB (Mean = 2.57), Durgapur (Mean = 2.57), and Alwar (Mean = 2.39) face relatively higher challenges, suggesting improvements that would be required in structured review mechanisms, clear KPI-based evaluations, and feedback loops that could enhance overall depot performance management.
- Smaller locations like Vijayawada (Mean = 2.14), Pantnagar (Mean = 2.11), and Bhandara (Mean = 1.79) show fewer challenges, indicative of effective localised performance management or fewer operational complexities, although we must note that the number of respondents from these areas are only a few as compared to others.

4.5: Financial and Cost Control Challenges

4.5.1 Overall Assessment of Financial Challenges

The mean score across financial and cost-control-related questions (Q34–Q39) is 2.94, indicating moderate-to-slightly high financial management challenges in the relationship with 3PL providers. Issues identified here may be significant enough to necessitate immediate managerial focus, particularly in terms of cost stability and predictability.

4.5.2 Significant Financial Challenges (Higher Scores)

Key financial challenges highlighted are:

- High logistics costs offsetting outsourcing benefits (Q36, Mean = 3.49): High mean scores indicate substantial concern among respondents regarding cost-benefit balance, suggesting the financial efficiency of outsourcing logistics may be deemed as not optimal.
- Cost fluctuations affecting delivery planning and budgeting (Q35, Mean = 3.41): Indicates that cost instability frequently may often hinder financial planning and may also significantly affect operational predictability as well as the budgeting processes.
- Emergency or unscheduled deliveries may result in incurring cost premiums (Q38, Mean = 3.35): Thereby possibly suggesting that unscheduled or urgent deliveries consistently cause inflated costs, which may pose financial planning and budgeting difficulties.

4.5.3 Moderate Financial Challenges (Intermediate Scores)

- Predictability and stability of freight rates (Q34, Mean = 2.73): Indicates that there are moderate challenges associated with maintaining stable and predictable freight rates, that could reflect intermittent volatility with regards to pricing.
- Transparency in invoicing (Q37, Mean = 2.37): Reflects relatively lower but still noteworthy challenges revolving transparency, which can lead to manageable but slightly recurring administrative complications as well.

4.5.4 Lower Financial Challenges (Lowest Score)

- Clarity in contractual terms regarding pricing (Q39, Mean = 2.29): The lowest mean score in this section reflects that contractual clarity on pricing is reasonably deemed to be good, indicating fewer misunderstandings or contractual disputes regarding pricing terms that are set.

4.5.5 Correlation with Overall Satisfaction (Q7)

- Cost fluctuations (Q35, correlation = -0.21) and emergency delivery premiums (Q38, correlation = -0.16) have noticeable negative correlations with satisfaction, thereby highlighting that unpredictable or inflated costs notably cause a decrease in overall satisfaction levels.
- Remaining items (Q34, Q36, Q37, Q39) have minimal correlations, suggesting these financial aspects, though challenging, have a less direct effect on general satisfaction.

4.5.6 Depot-Level Financial Challenge Analysis

- Depots with Elevated Financial Challenges:
    - Durgapur (Mean = 3.83) reflects the highest challenge levels, particularly emphasizing urgent and high-cost premium issues and relating unpredictability although we must keep in mind that there is less representation from there.
    - Pantnagar (Mean = 2.87) and ALCOB (Mean = 2.96) also face pronounced financial challenges, especially regarding cost fluctuations and logistics cost-benefit ratios.

- Depots with Relatively Lower Financial Challenges:
    - Vijayawada (Mean = 2.42), Pune (Mean = 2.75), and Ennore (Mean = 2.85) indicate fewer but sometimes prevailing financial-related issues, suggesting relatively stable cost management practices and slightly lesser exposure to volatile cost conditions that exist.

4.6: Strategic Fit and Relationship Management

4.6.1 Overall Strategic Fit and Relationship Management

The average mean score for strategic fit and relationship management challenges (Q40–Q45) is quite low at 2.09, indicating generally low-to-moderate levels of concern. Respondents report relatively effective alignment and relationship management practices between Ashok Leyland and the 3PL providers that they interact with.

4.6.2 Key Strategic and Relationship Management Challenges (Higher Scores)

The analysis identifies moderate challenges primarily in the following areas:

- Viewing 3PLs as strategic partners (Q45: Mean = 2.04) and long-term collaboration beyond cost (Q41: Mean = 2.14): Low scores indicate strong recognition of strategic relationships and collaborative partnerships with 3PLs beyond mere transactional engagements, which is a positive sign for maintaining long-term relationships, although more improvements can be made to make then 3PL service providers feel like an integral stakeholder.
- 3PL partners' understanding of long-term logistics goals (Q40: Mean = 2.20): Slightly elevated scores as compared to other metrics could reflect occasional misalignments, highlighting potential room for clearer communication and strategic engagement with logistics partners regarding Ashok Leyland's long-term objectives.
- Trust improvement over time (Q42: Mean = 2.12): Indicates slightly moderate challenges in consistently enhancing trust, suggesting trust levels have remained stable and could have a slight room for improvement.

4.6.3 Areas Demonstrating Relatively Stronger Strategic Alignment (Lowest Scores)

- Internal alignment with 3PL performance expectations (Q44: Mean = 2.00) and Ashok Leyland's training/support for 3PL partners (Q43: Mean = 1.96): Both items demonstrate relatively lower challenge scores, indicating that internal alignment and support mechanisms are effective and well-maintained.

4.6.4 Correlation with Overall Satisfaction (Q7)

Certain correlations reveal insights on strategic fit's influence on overall satisfaction:

- 3PL partners' understanding of long-term logistics goals (Q40: correlation = -0.28) and relationships based on long-term collaboration (Q41: correlation = -0.18) both show moderate negative correlations, indicating that clearer strategic alignment and collaboration directly enhance satisfaction levels and improve operational efficiency in the long-run.
- Other correlations (Q42–Q45) remain minimal, indicating these strategic aspects, while important, have less direct influence on immediate satisfaction outcomes regarding this aspect.

4.6.5 Depot-Level Strategic Relationship Analysis

- Durgapur (Mean = 3.67) shows notably higher strategic alignment challenges, suggesting significant improvement opportunities, particularly in terms of garnering partner trust and familiarising them with long term goals.
- ALCOB (Mean = 2.40) and Alwar (Mean = 2.33) have slightly elevated scores, indicating moderate alignment and relationship management challenges around strategic engagement and trust-building activities, although not entirely severe.
- Bhandara (Mean = 1.33), Hosur (Mean = 1.76), and Ennore (Mean = 1.76) report fewer strategic alignment issues, reflecting strong, strategically coherent practices and effective relationship management which are prevalent across these locations.

4.7: Challenge Ranking and Risk Perception

4.7.1 Ranking of Challenges (Severity Analysis)

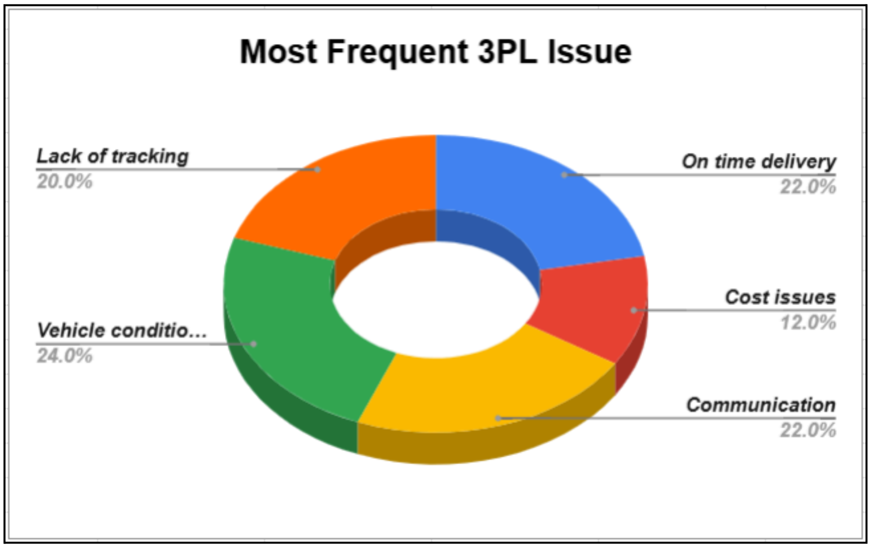

Figure 4.3: Most Frequent 3PL Issue as faced by Managers

Respondents were asked to rank specific challenges in terms of its severity (with higher scores indicating a greater perceived severity of one aspect over another):

- Delivery delays had emerged as the highest-rated issue overall (Mean = 3.67), which strongly suggests that delays are perceived as the most critical operational challenge that affect the outbound logistics within the organisation.
- Cost fluctuations (Mean = 3.20) are also significant, indicating that unpredictable costs remain a comparatively significant pain point, which impacts both planning as well as budgeting effectiveness.
- Vehicle transit damage (Mean = 2.94) as well as poor technology use (Mean = 2.79) also present moderate concerns, which reflects recurrent but relatively less critical problems compared to delays and costs.
- Communication gaps (Mean = 2.49) have been ranked as relatively less severe issues overall, thereby indicating a lower perceived impact compared to operational delays or cost-related issues.

4.7.2 Depot-Level Challenge Severity

- High-Severity Locations:

  - Ennore records notably high concern levels when it comes to delivery delays (Mean = 4.18) and cost fluctuations (Mean = 3.10).
  - Hosur and Pantnagar also rank delivery delays high (Mean = 3.67 each), thereby emphasising consistent urgency in managing delivery timelines across key operational zones.

- Moderate-Severity Locations:

  - ALCOB highlights transit damage (Mean = 3.63) and delivery delays (Mean = 3.63) as leading challenges too.
  - Alwar reflects substantial issues around communication gaps (Mean = 3.14) and cost fluctuations (Mean = 3.29).

- Lower-Severity Locations:

  - Vijayawada reports Transit Damage as the highest (Mean=3.5) with other factors remaining more or less the same, suggesting manageable logistics coordination at this smaller depot.

4.7.3 Most Frequent 3PL Issues (Respondent Perceptions)

When asked explicitly about the most frequent operational issues:

- On-time delivery (22%) and communication (22%) are jointly highlighted as one the most recurring issues, which makes them consistent with previous findings emphasising delays and inconsistent information flow.
- Vehicle condition and damages (24%) rank slightly higher and take the top spot as the most frequent issue, therefore underlining persistent challenges in maintaining vehicle condition standards during transit.

- Lack of tracking (20%) emerges as a considerable concern as well, reinforcing the earlier emphasis on digital tool adoption and transparency.
- Cost issues (12%), though perceived as highly impactful in severity rankings, are less frequently encountered day-to-day compared to operational and communication issues.

4.7.4 Risk Perception Trends

When asked whether 3PL-related risks have increased over time:

- 39.2% affirm rising risks, indicating substantial concern regarding future operational reliability.
- 35.3% are uncertain, signaling possible gaps in visibility or clarity regarding evolving logistics conditions and atmosphere.
- Only 25.5% believe risks are not increasing, suggesting that overall, most logistics managers sense an ongoing struggle or emerging challenges in managing 3PL partners.

4.8: Outbound Delivery Specific Challenges

4.8.1 Overview of Outbound Delivery Challenges

The average score for outbound-specific delivery challenges (Q51–Q54) is 2.94, highlighting moderate but critical challenges which could be directly attributed to vehicle dispatch, dealer interactions, and delivery quality.

4.8.2 Most Critical Outbound Delivery Issues (Higher Scores)

Respondents highlighted the following critical outbound logistics challenges:

- Delivery to remote regions within the country thereby facing frequent 3PL failures (Q53: Mean = 3.20): High scores strongly indicate pretty higher operational and logistical challenges in remote or geographically challenging regions. Such recurring failures directly affect service reliability and customer satisfaction in these areas.
- Vehicle condition subpar at the delivery point (cleanliness, damages) (Q54: Mean = 3.20): Equally high scores reflect frequent dissatisfaction with vehicle conditions upon

delivery, suggesting considerable lapses during vehicle handling, loading, or at the stage of transporting.

- Slight vehicle rework or rejection at dealer yards (Q52: Mean = 2.80): Moderately high scores emphasize issues involving vehicle quality checks at dealer yards that lead to additional rework efforts, delays, and increased operational costs as well.

4.8.3 Less Severe Outbound Delivery Issues (Lower Scores)

- Outbound deliveries delayed beyond expected timelines (Q51: Mean = 2.57): The lower relative mean indicates somewhat fewer challenges here are experienced as compared to other operational concerns in this section, although delays are still impactful enough to warrant continued attention in order to constantly keep improving.

4.8.4 Correlation with Overall Satisfaction (Q7)

- Delivery to remote regions (Q53: correlation = -0.10) has the strongest negative correlation, thereby indicating that recurrent failures in remote-region logistics negatively affect overall satisfaction.

- Other issues only show negligible correlations, suggesting they individually have limited direct influence on general satisfaction, though operationally impactful.

4.8.5 Depot-Level Analysis of Outbound Delivery Challenges

- Depots with Highest Outbound Challenges:

    - ALCOB (Mean = 3.16) reports significant issues, especially with regards to remote-region deliveries (Mean = 4.25) and vehicle conditions at delivery (Mean = 3.13).
    - Durgapur (Mean = 3) reports really high issues with regards to delivery to remote areas and a subpar condition at which the vehicles are delivered. Meanwhile Pune also reports higher issues overall (Mean = 3.75) with almost all the metrics being classified as high profile.

    - Alwar (Mean = 3.36) and Pantnagar (Mean =3.61) highlights substantial challenges, particularly regarding vehicle conditions (Mean = 3.71) as well as remote-area logistics (Mean = 3.57).
    - Ennore (Mean = 2.65) reports moderate yet significant challenges, especially with frequent rework at dealer yards (Mean = 2.73).

- Depots with Moderate to Lower Challenges:

    - Vijayawada (Mean = 2.50), and Hosur (Mean = 2.47) comparatively reflect lower or moderate levels of operational challenges, indicating relatively better-managed outbound logistics practices or fewer complex operational dynamics overall.

4.9: Comparative and Future Outlook – Analysis

4.9.1 Perceived Change in 3PL Performance Over Time

When asked to assess how 3PL performance has changed over the past two years and if it has seen an improvement, the most common response was a moderate improvement:

- 49% of respondents rated it a score of 4
- 25.5% rated it 3 (no significant change)
- 15.7% gave it a full 5 (strong improvement)
- A smaller share of respondents (9.8%) rated it 1 or 2, indicating perceived deterioration

This question is also not reverse code, so higher values mean better.

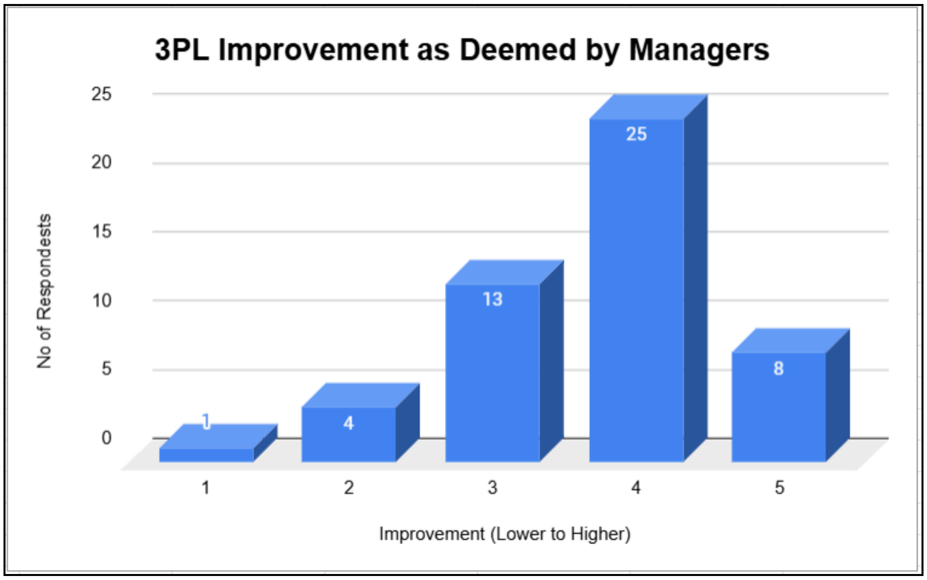

Figure 4.4: 3PL Improvement as Deemed by Managers

The overall average score is 3.69, suggesting that while significant transformation is not yet visible, there has been a clear upward trend in service quality.

Location-Wise Insights:

- Ennore (4.09) and Hosur (4.00) report the highest perceived improvement, indicating that vendors at these sites have likely responded well to expectations and may be better integrated into the operational processes as well.
- Pantnagar (3.56), ALCOB (3.38), and Alwar (3.57) report moderately positive change as well regarding this aspect.
- Durgapur, however, reports a sharp deviation with a low score of 2.00, possibly reflecting dissatisfaction based on localised issues, although it's based on a very limited sample size and should be interpreted cautiously and with more research required for that particular location.

These responses suggest that while most plants are seeing improvement, the progress is not completely uniformly distributed, and some regions may still be grappling with legacy vendor challenges or infrastructure gaps, which can be identified through further research.

4.9.2 Digital Transformation as a Solution

A significant majority of 72.5% believe that digital transformation can resolve a lot of 3PL challenges at present. Only 9.8% of the respondents had disagreed, with the remaining 17.6% uncertain or not sure about the whole situation.

This strong consensus reflects clear confidence in digital tools as a strategic enabler and an improver. Given the earlier findings on underutilised dashboards, weak integration, and reliance on manual records, it is evident that respondents see digital transformation not just as an IT upgrade, but as a pathway to operational consistency, transparency, agility and time saving as well.

4.9.3 Strong Support for Centralized 3PL Coordination

An overwhelming 86.3% of respondents favour centralising 3PL coordination, while just 13.7% prefer to keep this system decentralized.

This sentiment suggests that current decentralized coordination models are perhaps scattered across locations, teams, or systems and are seen as inefficient. Centralisation is viewed as a means to bring uniform accountability, standard performance reviews, and integrated technology platforms under one governance structure.

4.9.4 Interest in Hybrid Logistics Models

When asked if they would prefer a hybrid model (mix of in-house and outsourced logistics) over a fully outsourced 3PL model:

- 80.4% responded to a 'Yes'
- 19.6% preferred sticking with full 3PL outsourcing in these aspects

This reflects a growing realisation that exclusive reliance on 3PLs is insufficient, especially in areas requiring greater control over quality, speed, or cost predictability. A hybrid model would allow Ashok Leyland to retain operational control over critical routes or functions while leveraging 3PL flexibility in other dynamics

4.9.5 Key Areas for Improvement

Respondents were asked to prioritize the most urgent area for improvement in 3PL relationships:

- Technology dominated with a staggering 52.9%
- Training followed also at a significant 21.6%
- Cost structure, communication, and infrastructure were each selected by only 9.8% of the respondents.

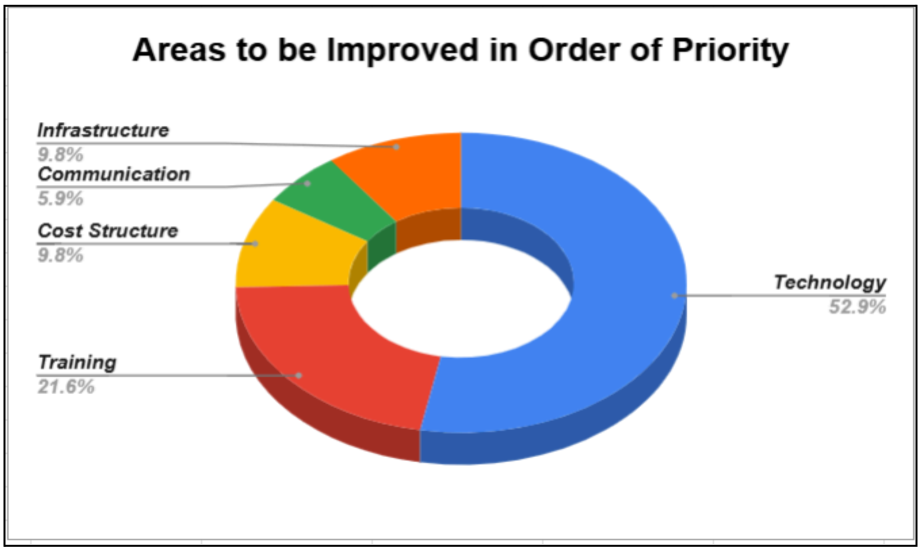

Figure 4.5: Areas to be Improved in Order of Priority

This again reinforces the earlier observations: the primary bottleneck is not pricing or physical infrastructure, but rather the absence of real-time systems, visibility tools, and digital planning platforms. Training is the next priority, suggesting that once tools are deployed, people need the know-how to use them effectively, as well as in more finer aspects such as the deployment of drivers to tougher terrains in order to mitigate accidents and reduce damages of the company's important product which are the vehicles.

4.9.6 Views on 4PL Potential

In response to the idea of implementing a 4PL model, wherein a single, integrated logistics partner would manage all 3PL providers, most of the respondents showed moderate optimism:

- 41.2% rated the potential as 3 (neutral to moderate promise)
- 25.5% rated it 4, and 17.6% rated it 5 (strong potential)
- A small group (15.7%) expressed skepticism, rating it at only a 1 or a 2

The feedback shows interest but also caution. While 4PL models offer strategic integration, respondents may be unsure of whether such a system would work in India's fragmented logistics landscape, or whether existing 3PLs can be unified under a single coordinating entity as well.

4.9.7 Future Growth of 4PL in the Heavy Vehicle Sector

Asked if the scope of 4PL in India's heavy vehicle logistics will increase significantly over the next five years:

- 41.2% agreed strongly (rating 4)
- 37.3% were moderately optimistic (rating 3)
- Only 11.8% expressed low confidence (ratings 1 or 2)

This indicates a positive long-term outlook, with over 78% of respondents expecting meaningful growth in integrated logistics solutions. Respondents see 4PL as a likely evolution in the industry, particularly to handle rising complexity in outbound vehicle distribution. However, this growth is seen as perceived only over a longer period of time.

4.10: Transit-Related Challenges

4.10.1 Overall Assessment of Transit Issues

With a grand average of 3.76 across 14 checked parameters (Q62–Q75), this section clearly reflects high-intensity logistical and operational concerns in the outbound delivery process when we look at issues through the micro-lens. This is a high scoring section which suggests that on-road and dispatch-related transit issues are the most acute and consistent problem area for the logistics teams.

4.10.2 Most Severe Transit Challenges (Top Mean Scores)

- Driver shortages during month-end billing cycles (Q62: Mean = 4.27): This has been reported as the most significant challenge faced across all locations. Month-end surge periods also lead to acute driver availability crises, potentially causing cascading delays and service quality issues.
- Rust cleaning, badging, and body builder readiness (Q64: Mean = 3.84, Q65: Mean = 4.12): High scores therefore indicate coordination gaps between vehicle readiness stages and dispatch timelines. These non-logistics delays are also clearly impacting overall outbound delivery efficiency.
- City “no-entry” time restrictions (Q70: Mean = 3.86) and RTO/border clearance (Q68: Mean = 3.59): These also show comparatively high systemic hurdles that reduce delivery flexibility and increase operational costs or overnight layovers.
- Inwarding delays due to space constraints or closures (Q73 & Q74: Means = 4.08 & 4.17 respectively): These are highly significant but somewhat external. However, they indicate a misalignment in scheduling and visibility between Ashok Leyland and its associated dealers.

4.10.3 Other Notable Challenges

- Vehicle breakdowns (Q67: Mean = 3.52) and driver health issues (Q72: Mean = 3.61): Reflect systemic concerns that surround fleet quality and transporter preparedness.
- Fastag deactivation by transporters (Q75: Mean = 3.59): Highlights increasing administrative or financial lapses from 3PL partners, leading to additional halts as well as legal complexities.

4.10.4 Correlation with Overall Satisfaction (Q7)

- Driver shortages during month-end billing (Q62: correlation = 0.36) has the strongest positive correlation with dissatisfaction, reinforcing that this is a major recurring obstacle that impacts overall satisfaction.

- Inwarding delays (Q73 & Q74) and route-level issues like city entry timing (Q70) also show moderate positive correlations (0.25–0.26 range), further validating their impact on user perception as well as logistics success.

4.10.5 Depot-Level Transit Risk Profiles

- High-Challenge Locations:

  - Pune (Avg = 4.43) and Pantnagar (Avg = 4.06) show widespread issues across all 14 transit parameters, thereby indicating systemic bottlenecks or inadequate route-level controls.
  - ALCOB, Ennore, and Alwar (Avg ≈ 4.0) also report intense levels of challenge, particularly related to coordination, driver availability, and entry restrictions.

- Moderate Challenge Zones:

  - Hosur (Avg = 3.72) and Bhandara (Avg = 3.50) reflect relatively more controlled environments, though challenges still persist.

- Lowest Challenge Zone:

  - Vijayawada (Avg = 3.43) shows comparatively fewer issues. This may reflect smaller operational volumes or better vendors in the regional locality. However, there is still a scope to enhance the overall performance.

Chapter 5: Strategic Managerial Recommendations

This chapter presents a targeted set of recommendations which are derived directly from the empirical evidence uncovered with the help of the quantitative analysis of Ashok Leyland's outbound logistics ecosystem. The intent is to move beyond diagnosis into actionable strategy, keeping in mind plant-specific nuances, stakeholder sentiments, and the future-readiness of logistics operations as well.

5.1 Strengthening Delivery Timeliness and Driver Management

- Addressing Peak-Period Driver Shortages: With Q62 scoring the highest (Mean = 4.27), it is crucial to deploy contingency driver rosters or pre-scheduled buffer arrangements, especially for month-end cycles. An incentive-driven pool of standby drivers can be developed in coordination with the transporters.
- Improving Adherence to Delivery Timelines: Depot-specific issues at Ennore, Pantnagar, and Hosur (with average challenge scores exceeding 3.5) necessitate tighter SLA enforcement with 3PLs and real-time dispatch scheduling that truly integrates plant readiness (Q64/Q65) and entry restrictions as well (Q70).

5.2 Enhancing Digital Maturity and Information Flow

- Combat Informal Communication Practices: With Q24 (Mean = 3.63) showing heavy dependence on informal channels, it is essential to implement plant-wide usage of structured communication tools (e.g., Transport Management Systems with mobile extensions), that will truly synchronise the entire process.
- Dashboards and Real-Time Tracking: Though digital dashboards (Q26, Mean = 1.96) are being utilised, deeper integration with driver location feeds, breakdown updates, and delivery status alerts should be enforced, particularly in Pune, Durgapur, and ALCOB which show elevated digital challenges.

5.3 Centralising 3PL Governance Structure

- With 86.3% of respondents favouring centralised coordination, it is recommended to:
  - Create a unified 3PL Control Tower that standardises performance reviews, SOP updates, and compliance feedback across all locations.
  - Deploy a central logistics operations dashboard visible to both corporate and depot-level managers to ensure alignment and early issue detections that may arise.

5.4 Route Planning and Remote Delivery Preparedness

- Route Pre-Planning Mechanism: Q13 (Mean = 2.98) reflects moderately low proactive planning which may happen at the part of the vendors. Also Integrating AI-supported route mapping or atleast using it as a support/reference and enabling pre-approved route grids for high-risk geographies could help reduce avoidable delays as well.
- Remote Area Logistics Reinforcement: With Q53 (Mean = 3.20), it is clear that deliveries to rural and far-off dealers are failing slightly more. Possible fixes could include, evaluating zone-based micro-warehouses or utilising last-mile specialist vendors for difficult regions.

5.5 Optimising Cost and Financial Control

- Emergency Delivery Cost Oversight: Higher means for Q38 (Mean = 3.35) and Q36 (Mean = 3.49) call for a stricter approval matrix for urgent dispatches and tighter contractual penalties for last-minute surges that may arise.

5.6 Building Long-Term Vendor Relationships and 4PL Feasibility

- 3PL Strategic Integration Programs: While strategic fit scored well overall, depot-level divergence (e.g., Durgapur: Mean = 3.67) reveals the need for structured partner onboarding, which could include strengthening SOP training and exposure to Ashok Leyland's long-term goals as well.
- 4PL Exploration Roadmap:
  - 41.2% of respondents see strong future potential.

- Begin a pilot engagement with one lead logistics integrator (LLI) to manage selected high-volume lanes, integrating multiple 3PLs under a common governance and visibility system as a test, in order to identify if any roadways are being made.

5.7 Depot-Specific Interventions

Depot	Key Issues Identified	Recommended Interventions
Ennore	Higher scores in driver route violations and over-speeding; faces challenges in SOP adherence and limited proactive route planning (Q13).	Strengthen route-planning procedures; reinforce SOP drills; introduce route simulation workshops for drivers.
Hosur	High perceived improvement; issues in delivery delays, emergency transport cost (Q36), and remote delivery challenges (Q53).	Apply penalty tracking for emergency dispatches; optimise cost controls; identify and assign drivers for remote areas.
Alwar	Problems in information flow (Q24), driver training (Q54), delivery timelines (Q64), and low dashboard usage.	Implement structured communication SOPs; conduct refresher training; assign depot-level digital champions.
Pantnagar	High challenge scores in SOP compliance, outbound delays, accident risks (Q75), and driver fatigue (Q56).	Introduce fatigue monitoring protocols; restructure night-dispatch SOPs; initiate HQ-level oversight of risky lanes.
ALCOB	Issues in vehicle condition on delivery (Q42), informal coordination, emergency costs (Q38), and dashboard underutilisation.	Enforce vendor-side inspections; link delivery conditions to scorecards; reinforce digital dashboard accountability.
Durgapur	Highest challenge means in strategic alignment (Q41), route-level execution, and SOP discipline; sample size limited.	Flag for zonal revalidation; initiate engagement audits with 3PLs; consider digital onboarding for all vendor staff.
Pune	Challenges in route planning (Q13), border delays (Q57), emergency cost increases, and SOP adherence variance among 3PLs.	Create escalation matrix for SOP breaches; digitise route SOPs; introduce monthly transporter performance review cycles.

Vijayawada	Moderate scores in SOP implementation, fatigue, and delivery delays; digital visibility and route mapping usage relatively underleveraged.	Introduce driver rotation planning; digitise tracking touchpoints; formalise escalation response SOPs with transporters.
Bhandara	Challenges noted in emergency cost management (Q36), SOP inconsistencies, and coordination gaps during peak periods.	Optimise emergency freight approval layers; strengthen coordination cadence with transporters for time-sensitive loads.

5.8 Building Hybrid Capacity and Internal Logistics Muscle

- With 80.4% preferring hybrid over fully outsourced logistics, Ashok Leyland can:
    - Identify high-sensitivity or problematic routes to be taken in-house (e.g., remote regions, high-accident corridors). Over-time the cost can be recovered as we can expect lesser accidents and vehicle damages. Further, this can boost on time sales leading to satisfaction on the part of both the dealers and the customers.
    - Invest in a trial fleet of owned transport units, especially from high-performing depots like Hosur and Ennore, to compare performance and cost structures side-by-side.

5.9 Training and Capability Building

- As 21.6% of respondents cited training as the next best priority, an annual training calendar should be designed for:
    - Driver SOP compliance
    - Digital tool usage across all plant logistics teams
    - Emergency escalation and disruption response training.

5.10 Final Takeaway

This analysis and its corresponding managerial roadmap reveal a logistics ecosystem that is operationally competent but riddled with coordination, technological, and compliance inconsistencies which if improved can lead to operational excellence and stakeholder satisfaction. Ashok Leyland stands at a pivotal moment to strengthen its outbound operations through:

- A combination of digital reform,
- Structural coordination models, and
- Depot-wise performance restructuring and compatriot benchmarking

Taking forward these recommendations with a system-wide approach, coupled with site-level ownership, will be critical in future-proofing the company's logistics network and reinforcing its leadership within the Indian heavy vehicle market.

Chapter 6: Limitations of the Report

While this study provides meaningful and data-driven insights into the challenges faced in Ashok Leyland's outbound 3PL coordination, it is important to recognize the boundaries and restrictions within which this research was conducted. The following limitations are acknowledged in order to contextualize the findings and inform future improvement opportunities in data collection, methodology, and strategic validation.

6.1 Sample Size and Representation Imbalance

Although the study achieved a satisfactory number of total respondents across functions and depots, the distribution was uneven across some plant locations. In particular:

- Locations such as Durgapur and Pune had much fewer respondents, which limits the reliability of their site-specific findings.
- Conversely, zones like Ennore and Hosur were more strongly represented, creating a heavier data influence from these locations

As such, depot-level comparisons should be interpreted with caution, especially when sample sizes are limited to one or two individuals, as mentioned before.

6.2 Exclusive Focus on Internal Stakeholders

The study is built exclusively on the inputs of Ashok Leyland's internal logistics personnel, across roles such as execution, planning, and location management. While this ensures operational authenticity, it also means:

- The perspectives of external 3PL providers, drivers, and transport coordinators were not captured directly and understood.
- Vendor-side operational constraints (e.g., manpower shortages, asset availability, regulatory hurdles) may not be fully reflected in the challenge ratings.

This creates an opportunity for future studies to adopt a multi-stakeholder approach by incorporating supplier-side viewpoints as well.

6.3 Static Quantitative Design

The survey used a structured, close-ended questionnaire with a fixed Likert-scale format. While this allows for statistical consistency and pattern analysis:

- It limits the depth of respondent explanation behind certain challenge ratings in order to come to more and better conclusions.
- Nuanced insights, such as how specific digital breakdowns occur or why a 3PL might resist centralisation could not be explored within the scope of this structure.

A mixed-method approach which could've included qualitative interviews or focus groups may have helped enhance depth in future research iterations.

6.4 Time-Bound Perception Bias

The data was collected during a specific period in 2025, which may reflect short-term conditions such as:

- Temporary policy changes at plant gates, or recency bias
- Ongoing digital rollouts
- Month-end or festival-period workload surges that may have caused a change in normal dynamics.

These factors may have influenced respondents' answers, especially in domains like delivery reliability, emergency freight usage, and route planning consistency. Therefore, the dynamic nature of logistics must be considered when using these insights, in order to facilitate long-term decision-making.

6.5 Scope Constraints

The scope of this analysis is deliberately limited to outbound logistics and 3PL partner coordination. As such:

- Inbound logistics, line feeding, yard operations, and inventory holding functions were excluded, and only one question with regards to inbound delays was asked.

- Similarly, commercial contract design, cost benchmarking against industry peers, and market competitiveness were beyond the intended scope of this project.

These exclusions should be acknowledged when extrapolating insights beyond outbound vehicle movement.

6.6 Interpretation of Correlation Values

Correlation values between challenge areas and satisfaction levels were used to derive key impact drivers. However:

- These correlations reflect association and not causation.
- While they help identify key problem clusters, they do not prove direct influence on satisfaction outcomes without deeper multivariate testing methodologies.

Hence, correlations should be seen as directional guides, not deterministic conclusions.

6.7 Generalisability Beyond Current Vendor Pool

The study's results reflect existing 3PL partners working with Ashok Leyland at the time of data collection. As such:

- Findings may not fully apply to new or alternate vendors introduced in the future procurement cycles.
- Any structural or technological shifts in the vendor ecosystem (e.g., adoption of control towers or real-time telematics platforms) may alter the relevance of certain observations that are noted.

Conclusion

Despite these limitations, this study provides a valuable and empirically grounded assessment of outbound 3PL coordination across Ashok Leyland's operational landscape. The findings should be treated as a strategic baseline, with future efforts building on this foundation through broader scope, stakeholder inclusion, and iterative performance tracking as well.

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Annexure

Final Quantitative Questionnaire: Challenges in 3PL Coordination at Ashok Leyland

Section 1: Respondent Profile

1. Your current department/function:
   - ☑ Inbound Logistics
   - ☑ Outbound Logistics
   - ☑ Both
   - ☑ Other (please specify): ____________
2. Work location:
   - ☑ Hosur
   - ☑ Pantnagar
   - ☑ Ennore
   - ☑ Other: ____________
3. Years of experience in logistics/supply chain:
   - ☑ Less than 2 years
   - ☑ 2–5 years
   - ☑ 6–10 years
   - ☑ More than 10 years
4. Academic qualifications:
   - ☑ Diploma
   - ☑ Bachelor's Degree
   - ☑ Master's Degree
   - ☑ Other: ____________
5. What logistics activities are outsourced to 3PL providers? (Select all that apply)
   - ☑ Inbound Transportation
   - ☑ Outbound Transportation
   - ☑ Warehousing
   - ☑ Last-mile delivery
   - ☑ Export/Customs Handling

- ☑ Other: ____________

6. What percentage of your operations rely on 3PLs?
    - ☑ <25%
    - ☑ 25–50%
    - ☑ 51–75%
    - ☑ >75%
7. Overall satisfaction with current 3PL providers:
    - ☑ 1 – Not at all satisfied
    - ☑ 2
    - ☑ 3
    - ☑ 4
    - ☑ 5 – Extremely satisfied

8. Which types of 3PL providers do you coordinate with most?

- ☑ Large national firms
- ☑ Regional SMEs
- ☑ Both

Section 2: Common Operational Challenges ***(Rate 1 = Strongly Disagree, 5 = Strongly Agree)***

9. 3PL partners frequently delay inbound deliveries.

10. Outbound vehicle deliveries often miss committed timelines.

11. Communication with 3PL teams is inconsistent.

12. 3PL service cost estimates often exceed initial quotations.

13. There is a lack of proactive route planning from 3PL providers.

14. Damages or losses during transit are common.

15. 3PL drivers do not consistently follow route/SOP compliance.

16. 3PL partners fail to accommodate urgent delivery requests.

17. Logistics disruptions are not quickly escalated or resolved.

18. SOP compliance by 3PLs is consistently followed.

19. There is clarity in responsibility ownership when errors occur.

Section 3: Technology and Digital Adoption

20. Which of the following tools are used by your 3PLs? (Select all)

- ☑ GPS Tracking
- ☑ Transport Management System (TMS)
- ☑ Electronic Proof of Delivery (ePoD)
- ☑ Manual records only
- ☑ Not sure

21. Rate the digital maturity of your 3PL partners:

- ☑ 1 – Very poor
- ☑ 2 – Poor
- ☑ 3 – Average
- ☑ 4 – Good
- ☑ 5 – Excellent

22. Real-time updates from 3PL partners improve operational efficiency.

23. Lack of digital tools delays coordination and decision-making.

24. Our team often relies on phone or WhatsApp updates instead of digital platforms.

25. Integration with our internal systems (ERP/TMS) is seamless.

26. Use of digital dashboards improves transparency with 3PLs.

Section 4: Performance Measurement & Review ***(Likert scale)***

27. 3PL performance is reviewed regularly based on clear KPIs.

28. Performance issues are addressed through structured reviews.

29. There is a feedback mechanism to report 3PL failures.

30. Delivery performance (timeliness, accuracy) is tracked and shared.

31. Penalties are enforced for repeated service failures.

32. 3PL performance has shown measurable improvement over the past 2 years.

33. 3PLs exceed expectations in key service areas.

Section 5: Financial and Cost Control Challenges ***(Likert scale)***

34. Freight rates by 3PLs are predictable and stable.

35. Cost fluctuations affect delivery planning and budgeting.

36. High logistics costs offset the benefits of outsourcing.

37. Transparent invoicing is maintained by 3PL providers.

38. Emergency or unscheduled deliveries incur significant cost premiums.

39. There is clarity in contractual terms regarding pricing.

Section 6: Strategic Fit and Relationship Management ***(Likert scale)***

40. 3PL partners understand our long-term logistics goals.

41. Relationships with 3PLs are based on long-term collaboration, not just cost.

42. Trust in 3PL partners has improved over time.

43. Ashok Leyland supports 3PL partners with training or integration.

44. Internal teams are aligned with 3PL teams on performance expectations.

45. We view 3PLs as strategic partners, not vendors.

Section 7: Challenge Ranking and Risk Perception

46. Rank the following in order of severity *(1 = Most Severe, 5 = Least Severe)*:

- Delivery Delays
- Poor Tech Use
- Cost Fluctuations
- Communication Gaps
- Transit Damage

47. What is the most frequent 3PL issue you face? (Select one)

- ☑ Timeliness
- ☑ Cost issues
- ☑ Communication
- ☑ Vehicle condition/damage
- ☑ Lack of tracking

48. Do you consider 3PL risks to be increasing in the last 2 years?

- ☑ Yes
- ☑ No
- ☑ Not sure

49. Have you recommended changing a 3PL provider due to recurring issues?

- ☑ Yes
- ☑ No

50. What is your confidence in 3PL partners during high-volume periods? *(1 = Very low, 5 = Very high)*

Section 8: Outbound Delivery Specific Challenges (Likert Scale)

51. Outbound deliveries to dealers are frequently delayed beyond expected timelines.
52. We face frequent vehicle rework or rejection issues at dealer yards.
53. Delivery to remote regions (e.g., Northeast, border areas) faces more 3PL failures.
54. Vehicle condition (cleanliness, scratches, fuel level) is often subpar at delivery point.

Section 9: Comparative and Future Outlook

55. Compared to 2 years ago, overall 3PL service quality has:

- ☑ Significantly worsened
- ☑ Slightly worsened
- ☑ No change
- ☑ Slightly improved
- ☑ Significantly improved

56. Do you think digital transformation will resolve major 3PL challenges?

- ☑ Yes
- ☑ No
- ☑ Not sure

57. Is there a need for centralizing 3PL coordination internally?

- ☑ Yes
- ☑ No

58. Would you prefer a hybrid 3PL-internal model over fully outsourced logistics?

- ☑ Yes
- ☑ No

59. What area should be prioritized for 3PL improvement? (Select one)

- ☑ Technology

- ☑ Training
- ☑ Cost structure
- ☑ Communication
- ☑ Infrastructure

60. The concept of 4PL (a single integrated logistics partner managing all 3PLs) holds strong potential for improving outbound logistics in the commercial vehicle industry. (Likert Scale)

61. Do you believe the scope of 4PL solutions in India's heavy vehicle sector will increase significantly in the next 5 years. (Likert Scale)

Section 10: Transit Related Challenges (Likert Scale)

62. Driver shortages during month-end billing cycles

63. Fund release/payment delays impacting dispatch

64. Delay due to badging or rust-cleaning requirements

65. Delays in body builder vehicle readiness confirmation

66. Driver shortages during major festivals (e.g., Deepavali, Ramzan)

67. Vehicle breakdown enroute

68. RTO/border tax clearance issues

69. Roadblocks due to festivals/yatras (e.g., Kavad, Maha Shivratri)

70. City "no-entry" time restrictions

71. Accidents during delivery

72. Driver health/emergency issues enroute

73. Inwarding delays due to space constraints at dealer

74. Inwarding delays due to Sunday/holiday closures

75. Fastag cancellation by transporters